USN | | | | | | | | | | [23TCSE403]

FOURTH SEMESTER B.Tech EXAMINATION, FEB -2025

THEORY OF COMPUTATION

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

			Blooms Level	CO	Marks
		PART-A	**5 x2=10**		
1.	**a.**	Define ambiguity in grammar with an example . .	L1	CO1	**02**
	b.	Illustrate the difference between string and Language.	L2	CO2	**02**
	c.	Describe DFA with all possible tuples.	L3	CO3	**02**
	d.	Explain PDA with tuples.	L1	CO2	**02**
	e.	Define Instantaneous description.	L1	CO5	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	Illustrate the difference between DFA and NFA.	L3	CO2	**05**
		OR			
	b.	Sketch a DFA to accept the strings of a's and b's ending with 'abb' and test the DFA by taking suitable string.	L2	CO2	**05**
3.	**a.**	Draw a DFA to accept the strings of a's and b's ending with 'ab' or 'ba' and define all the tuples of the machine.	L2	CO2	**05**
		OR			
	b.	Prove that L = $\{a^n b^n / n>=1\}$ is not regular using Pumping Lemma.	L3	CO3	**05**
4.	**a.**	Explain different types (Chomsky hierarchy) of grammars in detail.	L3	CO3	**05**
		OR			

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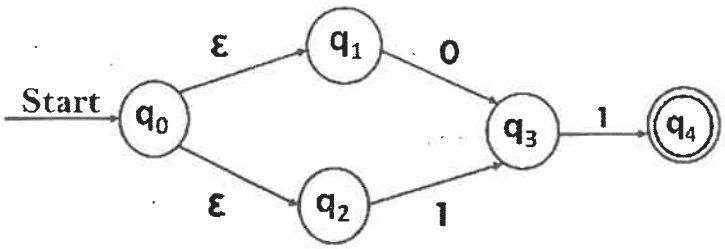
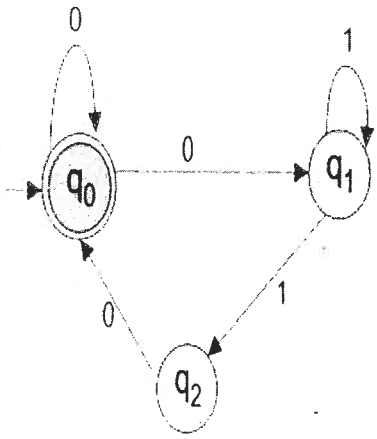
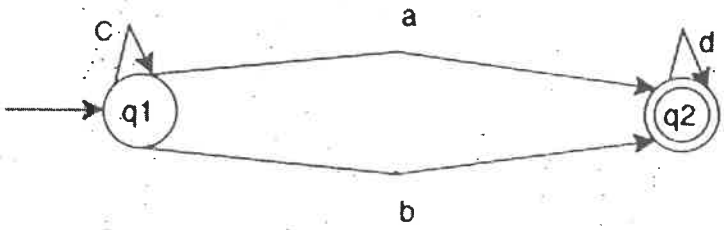
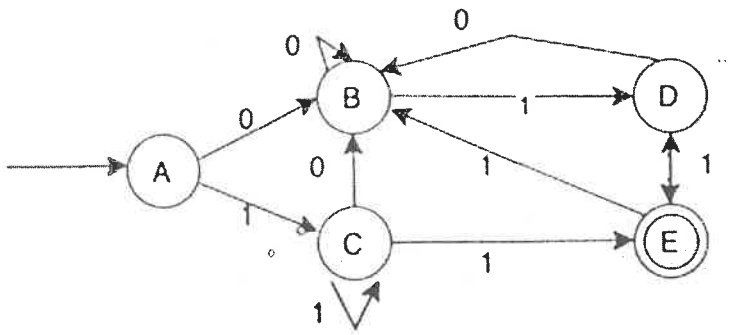
	b.	Define grammar and explain all the tuples with suitable examples.	L2	CO5	**05**
5.	**a.**	Explain all the properties of Regular Expression.	L2	CO5	**05**
		OR			
	b.	Obtain a NFA for the regular Expression $a^* + b^* + c^*$ with proper explanation.	L3	CO4	**05**
6.	**a.**	Define PDA with all the necessary parameters and mention it's uses and limitations.	L2	CO5	**05**
		OR			
	b.	What is two stack PDA ? Explain with suitable example.	L2	CO5	**05**
7.	**a.**	Design the Turing Machine to add two unary numbers.	L4	CO6	**05**
		OR			
	b.	Construct the Turing Machine for the language $L = \{ a^n b^n / n >= 1\}$ and write Instantaneous description for the same	L4	CO5	**05**
		PART-C	**6 X 10=60**		
8.	**a.**	Convert epsilon NFA to DFA 	L2	CO2	**10**
		OR			

	b.	Convert epsilon NFA to DFA 	L2	CO2	**10**
9.	**a.**	Convert FA to Regular Expression by using state elimination techniques 	L2	CO2	**10**
		OR			
	b.	Construct the Turing Machine to accept the both odd and even number of palindromes.	L2	CO2	**10**
10.	**a.**	Construct a PDA for the following CFG and test weather 'abbabb' is in N(P) G = ({S , A} , {a , b} , R , S) R = { S -> AA / a A -> SA / b }	L4	CO3	**10**
		OR			
	b.	Discuss the all closure properties of Regular Language with suitable example.	L3	CO3	**10**

11.	**a.**	Convert the below CFG to CNF. S - > bA / aB A -> bAA / aS / a B -> aBB / bS / b	L4	CO3	**10**
		OR			
	b.	What is recursion? Give an example of left and right recursion and explain how to eliminate the left recursion by taking suitable example.	L2	CO3	**10**
12.	**a.**	What is push down Automata? Sketch the pushdown automata for the language L = $\{0^n 1^n / n >= 1\}$.	L5	CO5	**10**
		OR			
	b.	Convert below given PDA to CFG. P= ({q0 , q1} , {a , b), {a , z0} , δ , q0 , z0) δ(q0 , a , z0) -> q0 , az0) δ(q0 , a , a) -> (q0 , aa) δ(q0 , b, a) -> (q1 , a) δ(q1 , b , a) -> (q1 , a) δ(q1 , a , a) -> (q1 , ε) δ(q1 , ε , z0) -> (q1 , ε)	L4	CO5	**10**
13.	**a.**	Construct the Turing machine for the following Language. L = $\{0^n 1^n 2^n / n>=1\}$	L4	CO5	**10**
		OR			
	b.	Minimise the below DFA by equivalence method.  	L4	CO5	**10**